Dengue Seroprevalence in Malaysian Adult:
A Comparison Study of Urban & Rural Population

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INTRODUCTION

As a dengue hyperendemic country, Malaysia monitors its dengue cases using the passive surveillance system. However, current system is only able to capture symptomatic cases. Since most dengue virus (DENV) infection are asymptomatic, periodic serosurvey is essential to determine the magnitude of DENV infection.

The Malaysian Cohort (TMC) has recruited at least 100,000 healthy Malaysian and their biological samples, such as serum, was stored in the biobank. This study utilized blood samples from TMC project and a total of 2958 samples were used in this dengue seroprevalence study.

METHODS

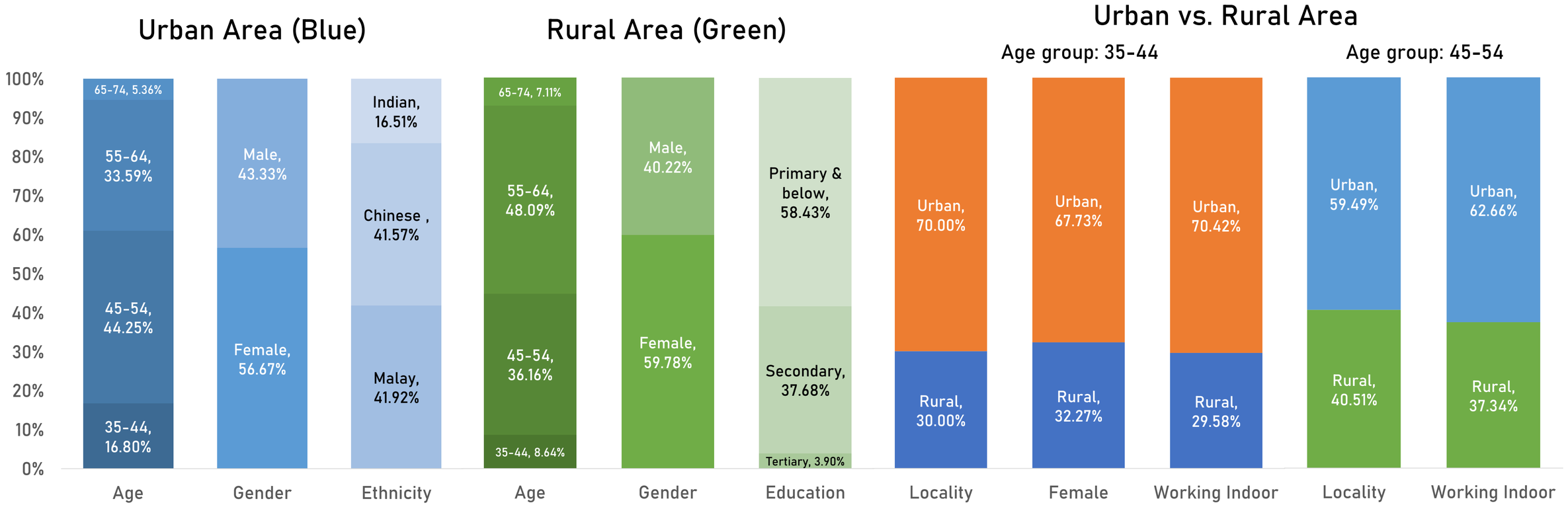
2 589 samples were retrieved from biobank of The Malaysian Cohort

- Enzyme-linked immunosorbent assay → To determine the history of DENV infection via immunoglobulin G, IgG
- Plaque reduction neutralization test assay → To determine neutralizing antibodies to DENV 1-4 in IgG positive samples
- Statistical analysis → To identify the significant associated risk factors

RESULTS AND DISCUSSION

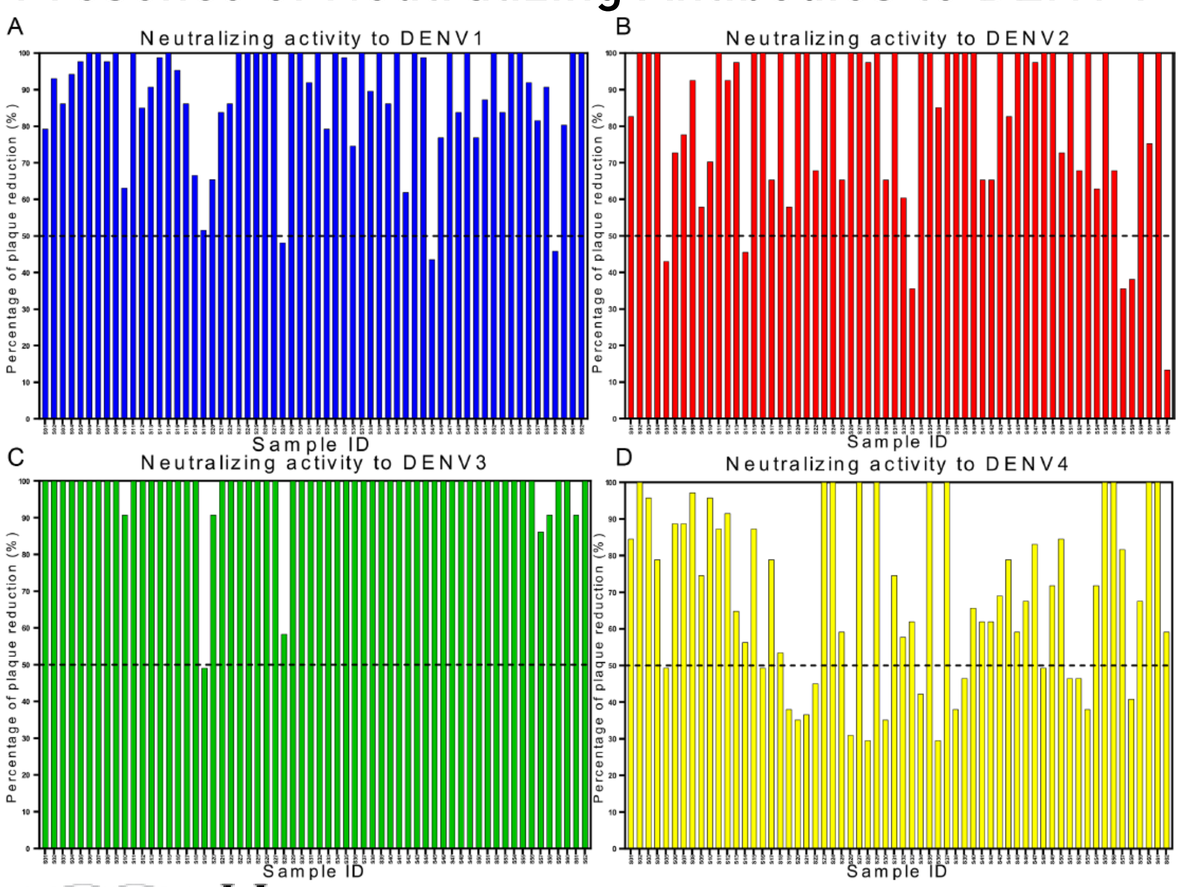
Dengue IgG Seroprevalence* in Adults of

*Only significant results are shown.

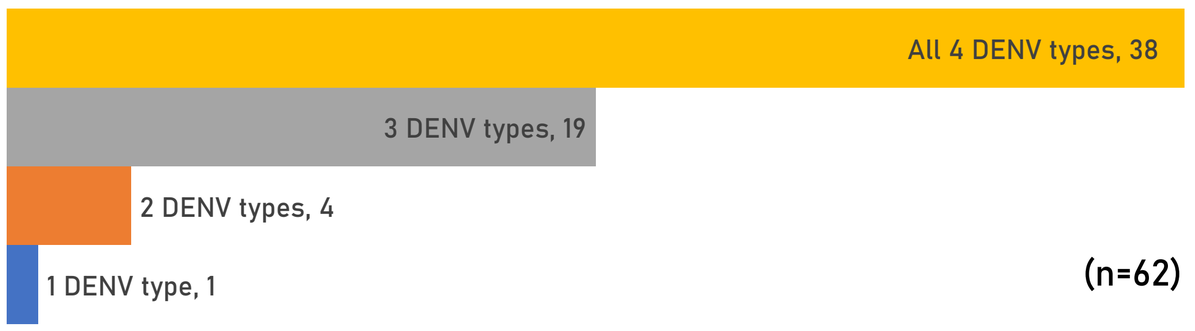


Adults in urban areas are more likely to have been exposed to DENV infection than those lived in rural areas.

Presence of Neutralizing Antibodies to DENV 1-4 in Dengue-IgG Positive Serum



In dengue IgG-positive serum samples, 98.39% of the samples had neutralizing antibodies against DENV3, but only 70.97% of them had neutralizing antibodies against DENV4.



(n=62)

Interestingly, as many as 61.3% of samples are having neutralizing antibodies against all 4 serotypes of DENV, indicating that they have been infected by all 4 types of DENV previously.

CONCLUSION

The high seroprevalence of dengue found in urban and rural areas suggests that both urban and rural communities are vital for establishing and sustaining DENV transmission in Malaysia.

REFERENCE

Azami, Nor Azila Muhammad, et al. "Dengue epidemic in Malaysia: urban versus rural comparison of dengue immunoglobulin G seroprevalence among Malaysian adults aged 35–74 years." *Transactions of The Royal Society of Tropical Medicine and Hygiene* (2020)..

Note: This poster was adapted by Ying-Xian Goh, Nor Azila Muhammad Azami, Mohd Arman Kamaruddin & Rahman Jamal for communication purpose only.

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